Reconstruction of the S-Wave Velocity via Mixture Density Networks With a New Rayleigh Wave Dispersion Function

Jianxun Yang, Chen Xu, and Ye Zhang

***Abstract*— How to determine the velocity of an S-wave from measured seismic data is an important topic in seismology. A modern technique for obtaining the S-wave velocity is to solve an inverse problem so that the simulated dispersion curve (the relation between the frequencies and phase velocities) coincides with the actual experimental results. In this work, by using the seismic impedance tensor, we propose a new function describing Rayleigh wave dispersion in the layered medium model of the Earth, which offers an efficient way to compute the dispersion curve. With this newly established forward model, based on mixture density networks (MDNs), we develop a physics-informed neural network, named MDN based on a physics informed forward model (FW-MDN), to estimate the S-wave velocity from dispersion curves. The FW-MDN method deals with the nonuniqueness issue encountered in the inversion of dispersion curves for the crust and upper mantle models, and attains satisfactory performance on an artificial dataset with various noise structures. Numerical simulations are performed to show that the FW-MDN offers easy calculation, efficient computation, and high precision for model characterization.**

***Index Terms*— Inverse problems, machine learning, mixture density network (MDN), Rayleigh wave dispersion.**

I. Introduction

THE study of surface waves has recently gained wide attention in seismic sounding since it plays an essential role in revealing Earth's structures from the shallow near-surface to several hundred kilometers deep into the mantle, depending on the frequencies and data acquisition configurations. As a kind of surface wave, the Rayleigh wave has a smaller velocity than that of the P-wave or the S-wave, which makes it more sensitive to changes in the underground medium through which it travels. Moreover, the energy attenuation of Rayleigh waves is slower than the geometrical divergence of body waves, which is strong and easy to identify in seismic records. Furthermore, the Rayleigh wave can be used to obtain information on the S-wave velocity structure in various mediums (e.g., the crust and the lithosphere). Hence, it can be considered the principal wave between a transverse and a longitudinal wave propagating along the Earth's surface. The Rayleigh wave has been widely applied in the modeling of small-depth geophysical sounding [1], [2], deep seismological studies [3], and other scientific areas [4].

Deriving Rayleigh wave dispersion curves from the crust and upper mantle models is an important item in the study of dispersion characteristics and surface wave sounding technology. The main goal of this topic is to establish an implicit equation regarding the relationship between the Rayleigh wave velocity and the wave frequency and other elastic parameters in the known medium model based on elastic wave theory to gain insight into the dispersion characteristics of the Rayleigh wave in the specific medium model. For a layered medium, there exist a large number of works in the literature. The classic propagator matrix method [5] is the first systematical and efficient method for calculating the velocity of surface waves and the corresponding eigenvalue problems. However, this approach is unstable. That is, the result obtained by this method is inaccurate in the case of high frequencies. To overcome this defect, many improved approaches, which are usually based on the propagator matrix method, have been developed, e.g., the Schwab–Knopoff method [6] and the Abo–Zena method [7], [8]. The idea of these methods is to improve the transfer matrix of the displacement stress vector to eliminate the exponential term in the transfer process and obtain accurate results. Another line of research, initiated by Kennett and Kerry [9] and named the reflection and transmission coefficient method, has recently gained considerable attention. Compared with the transfer matrix-based approaches, the reflection and transmission coefficient method is stable and very easy to implement in practice. However, when solving the dispersion equation in the reflection and transmission coefficient method, there exists the issue of a "mistake solution" [10]–[12]. To address this weakness, by combining the propagator matrix method and the reflection and transmission coefficient method, we propose a new Rayleigh surface wave dispersion forward model, which is shown in this work to be a robust method for estimating dispersion characteristics, such as the traveling surface wave propagation constant.

The inverse problem of estimating the medium parameters of the Earth from the measured characteristics of waves (e.g., the dispersion curve) is an important issue in geo-

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physics [11], [12]. Generally, numerical methods for solving these inverse problems are usually classified into three groups: linearization methods, heuristic global-optimization methods, and data-driven methods. The output of linearization methods is very sensitive to the initial choice of the solution and the accuracy of the computed Jacobian matrix, which is often obtained through the adjoint method [13]. Therefore, these methods often fail in real-world applications. The heuristic global-optimization methods, such as the annealing simulation algorithm [14], the particle-swarm global-optimization method [15], the genetic algorithm (GA) [16]–[18], and others [4], [19]–[21], are usually derivative-free methods, and their implementation is simple. Hence, during the last decade, they have been widely used in practice. The main defect of the heuristic global-optimization methods is their slow performance. Even for a small problem, they usually require several weeks to find the correct global minimizer.

Data-driven methods, especially the machine learning methods, address all the above problems and have begun to show potential in geoscience. For example, Liu *et al.* [22] propose a method that applies convolutional neural networks (CNNs) and the principal component analysis to parameterize complex geological models in low dimensions. Wang *et al.* [23] apply the CNN to reconstruct high-resolution porous structures in rock samples from low-resolution μ-CT images and high-resolution scanning electron microscopy images. Holm-Jensen and Hansen [24] address a linear probabilistic waveform inversion problem in cross-hole tomography with machine learning techniques. In their study, the forward model is approximated with a linear operator using the ridge regression algorithm, which helps them find an analytical solution to the inversion problem. Mosser *et al.* [25] use generative adversarial networks that represent prior distributions of geological heterogeneities in their study on the stochastic seismic waveform inversion problem. He and Wang [26] reparameterize model parameters in the full-waveform inversion problem as the weights of a deep neural network, which is known to have a strong representation capacity. For more recent progress on deep learning methods in geophysics, we refer to the latest review paper by Yu and Ma [27] and the references therein.

In the field of surface wave tomography, applications of machine learning are still limited, yet they have been proven to be successful. For example, Meier *et al.* [28] use the feed-forward neural network (FNN), and Cheng *et al.* [29] use stacked autoencoders to invert surface wave data for crustal thicknesses. Hu *et al.* [30] use CNNs to invert the theoretical Rayleigh wave phase and group velocity images for velocities. However, the models in these studies are unable to deal with a significant problem that is prevalent in surface wave tomography, i.e., the nonuniqueness problem (see [31] and [32]). That is, one dispersion curve may correspond to several distinct medium models, which leads to great difficulties in solving the inverse problem. The second part of our work contributes to solving these difficulties in a novel way by designing a new neural network architecture. The design is based on the mixture density network (MDN) model and the idea of physics-informed neural networks [33], [34] (the loss function is constructed using the forward model). The MDN was introduced by Bishop [35], [36] and is now widely used in various areas (e.g., it has been used in text generation [37], image recognition [38], unsupervised distribution and density estimations [39], and transportation forecasting [40]). We are the first to apply it in the field of surface wave tomography. However, note that the use of our model-design method is not restricted to tomography. It serves as an example for solving any type of inverse problem that requires finding multiple solutions for a single input.

TABLE I
NOTATIONS OF PHYSICAL QUANTITIES AND THEIR UNITS

Notation	Description	Unit
h	The thickness of the medium layer	Km
T	The period length	second (s)
f	The frequency	Hz
ω	$\omega = 2\pi f = \frac{2\pi}{T}$	rad/s
C	Rayleigh wave phase velocities	Km/s
γ	Rayleigh wave number (Rayleigh wave characteristic) $\gamma = \omega/C$	rad/Km
ρ	The density	Kg/m^3
λ, μ	Lame elastic constants	–
z	Earth's depth	Km
P-wave	Primary wave (pressure wave)	–
S-wave	Secondary wave (shear wave)	–
V_P	$V_P = \sqrt{(\lambda + 2\mu/\rho}$ – P-wave velocity	Km/s
V_S	$V_S = \sqrt{\mu/\rho}$ – S-wave velocity	Km/s
K_S	$K_S = \omega/V_S$ – S-wave number	rad/Km
K_P	$K_P = \omega/V_P$ – P-wave number	rad/Km
η_s	$\eta_s = \sqrt{\gamma^2 - k_s^2}$	–
η_p	$\eta_p = \sqrt{\gamma^2 - k_p^2}$	–
u_x	OX component of displacement for the Cartesian coordinate system OXYZ	–
u_z	OZ component of displacement for the Cartesian coordinate system OXYZ	–
σ_{xz}	The tangential stress	Pa
σ_{zz}	The normal stress	Pa

In sum, the main contributions of our work are twofold: 1) we develop a new mathematical model for efficiently computing the dispersion curves of Rayleigh waves, which avoids the pitfall of "mistake solutions" faced by conventional approaches and 2) we construct a new machine learning architecture [MDN based on a physics informed forward model (FW-MDN)] to solve the inverse problem—to invert Rayleigh wave dispersion curves for S-wave velocities—that contributes to solving a prevalent problem that cannot be solved by other machine learning models applied previously in surface wave tomography—the nonuniqueness problem. The technique used in our solution may also be helpful for other problems in the geosciences whenever finding multiple solutions for a single input is required. All codes, including the one for generating data, are publicly available on GitHub at https://github.com/chen-research/A-new-math-model-for-Rayleigh-wave-and-a-MDN-inversion-solver.

Notations: Table I summarizes the notations of some physical quantities that are frequently used in this article.

The remainder of this article is structured as follows. In Section II, we introduce a new recurrence equation for the dispersion of Rayleigh waves in a homogeneous and

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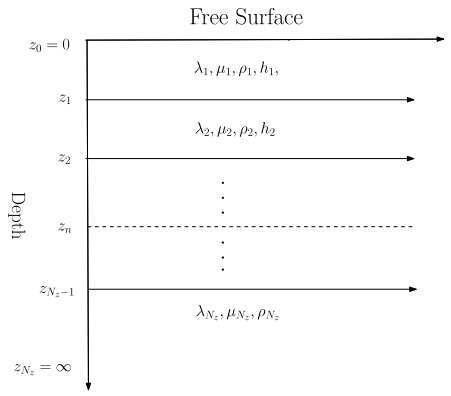


Fig. 1. Layered medium model of the Earth.

layered media. Moreover, in this section, we formulate the forward and inverse problems by using rigorous mathematical language. In Section III, based on the MDN, we develop a new machine learning method for solving our inverse problem in geophysics. A detailed construction procedure and numerical experiments of our machine learning method are also reported in Section III. Finally, concluding remarks are given in Section IV.

II. Novel Recurrence Equation for the Dispersion of Rayleigh Waves in Homogeneous and Layered Media

In this section, by introducing the seismic impedance tensor, we develop a new mathematical model for the efficient computation of the traveling wave characteristic in a layered medium. To that end, let us consider a layered medium with the boundary $z = z_n, n \in \{1, \ldots, N_z\}$, where $z_0 = 0$ represents the Earth's surface. In our simplified model (see Fig. 1), inside each layer, the Lamé coefficients λ and μ, as well as the Earth's density ρ, are all assumed to be constants. Specifically, for $z \in [z_{n-1}, z_n]$, $\lambda \equiv \lambda_n$, $\mu \equiv \mu_n$, and $\rho \equiv \rho_n$. Moreover, the Cartesian coordinate system $OXYZ$ is constructed such that the traveling surface wave propagates along the OX-axis in this medium. Then, the wave displacement vector $\vec{U}$ has the form

$$\vec{U}(x, y, z) = \overline{u}(z)e^{i\gamma x + i\omega t} \tag{1}$$

where $i = \sqrt{-1}$ denotes the imaginary unit, ω denotes the frequency of the seismic field, γ represents the traveling surface wave propagation constant (characteristic), and $\overline{u}(z) = (u_x, u_y, u_z)$ is the traveling surface wave amplitude. By the assumption in our simplified model, inside a layer (e.g., $z \in [z_{n-1}, z_n]$), $\vec{U} = (U_x, U_y, U_z)$ satisfies the Lamé equation with constant coefficients [see ([41])]

$$(\lambda_n + 2\mu_n)\text{grad}\,\text{div}\vec{U} - \mu_n \text{rot}\,\text{rot}\vec{U} + \omega^2\rho_n\vec{U} = 0 \qquad z \in [z_{n-1}, z_n]. \tag{2}$$

For Rayleigh waves, we need to consider only the OX and OZ components. By combining (2) and the stress equation ([7])

$$\sigma_{ij} = \lambda\delta_{ij}\text{div}\vec{U} + \mu\left(\frac{\partial U_i}{\partial j} + \frac{\partial U_j}{\partial i}\right), \quad \text{for all } i, j \in \{x, y, z\}$$

we have

$$\begin{cases} (\lambda_n + 2\mu_n)i\gamma\left(i\gamma u_x(z) + u_z'\right) - \mu_n\left(-u_x'' + i\gamma u_z'\right) \\ \qquad + \omega^2\rho_n u_x = 0 \\ (\lambda_n + 2\mu_n)\left(i\gamma u_x'(z) + u_z''\right) - i\gamma\mu_n\left(u_x' + i\gamma u_z\right) \\ \qquad + \omega^2\rho_n u_z = 0 \\ \sigma_{xz} = \mu_n\left(i\gamma u_z(z) + u_x'(x)\right) \\ \sigma_{zz} = (\lambda_n + 2\mu_n)u_z'(z) + i\gamma\lambda_n u_x(x) \end{cases} \tag{3}$$

where δ_{ij} denotes the Kronecker delta function and $(\cdot)' = (d/dz)(\cdot)$ refers to the derivative with respect to the variable z. The boundary conditions for (3) in layered media are given as follows.

1) On the Earth's surface, the stress tensor is zero

$$\sigma_{xz}|_{z=0} = \sigma_{zz}|_{z=0} = 0. \tag{4}$$

2) Displacements and tensions are continuous for the jth layer. That is,

$$\begin{aligned} u_x(z_j)|_{z=z_j+0} &= u_x(z_j)|_{z=z_j-0} \\ u_z(z_j)|_{z=z_j+0} &= u_z(z_j)|_{z=z_j-0} \\ \sigma_{xz}(z_j)|_{z=z_j+0} &= \sigma_{xz}(z_j)|_{z=z_j-0} \\ \sigma_{zz}(z_j)|_{z=z_j+0} &= \sigma_{zz}(z_j)|_{z=z_j-0}. \end{aligned} \tag{5}$$

3) The vanishing boundary condition is

$$\lim_{z\to\infty} u_x(z) = \lim_{z\to\infty} u_z(z) = 0. \tag{6}$$

In this work, we assume that the seismic impedance tensor of a Rayleigh wave links the stresses σ_{xz} and σ_{zz} with the displacements u_x and u_z by linear relationships

$$\begin{cases} \sigma_{xz} = Z_{xx}u_x + Z_{xz}u_z \\ \sigma_{zz} = Z_{zx}u_x + Z_{zz}u_z \end{cases} \tag{7}$$

where Z_{xx}, Z_{xz}, Z_{zx}, and Z_{zz} are four constants, which should be determined later. An explanation of (7) can be found in the work of Yang [18], [42] and Dmitriev and Akkuratov [41]. To this end, define the seismic impedance tensor $\hat{Z}$ of rank 2 by

$$\hat{Z} = \begin{pmatrix} Z_{xx} & Z_{xz} \\ Z_{zx} & Z_{zz} \end{pmatrix}. \tag{8}$$

By (3) and the continuation condition (5), one can conclude that the impedance tensor $\hat{Z}$ is continuous with respect to the parameters λ, μ, and ρ.

Considering the impedance tensor on the surface of the Earth, we derive from the boundary condition (4) for the stress (denoted as $\hat{Z}_0 = \hat{Z}(z = 0)$) that

$$\begin{aligned} 0 = \sigma_{xz}(z = 0) &= \overset{(0)}{Z}_{xx}u_x(z = 0) + \overset{(0)}{Z}_{xz}u_z(z = 0) \\ 0 = \sigma_{zz}(z = 0) &= \overset{(0)}{Z}_{zx}u_x(z = 0) + \overset{(0)}{Z}_{zz}u_z(z = 0). \end{aligned} \tag{9}$$

Hence, a Rayleigh wave exists (i.e., $u_x(0)u_z(0) \neq 0$) if and only if the following equation holds:

$$\det\hat{Z}_0 = \overset{(0)}{Z}_{xx}\overset{(0)}{Z}_{zz} - \overset{(0)}{Z}_{xz}\overset{(0)}{Z}_{zx} = 0. \tag{10}$$

The essential idea in this work is to use the dispersion (10) to establish the new relationship between the Rayleigh wave

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propagation constant γ and other parameters, such as the frequency ω and the parameters of the layered medium.

Denote by $u_s(z)$ and $u_p(z)$ the solenoidal and potential parts of the displacement field, respectively (see [5]–[7]). Then, for a Rayleigh wave, we have

$$\begin{aligned} u_x &= -u_s'(z) + i\gamma u_p(z) \\ u_z &= i\gamma u_s(z) + u_p'(z) \end{aligned} \tag{11}$$

where $u_s(z)$ is the solution of equation $u_s'' - \eta_s^2 u_s = 0$. Here, $\eta_s = (\gamma^2 - K_S^2)^{1/2}$, $k_s = (\omega/V_S)$, Re $\eta_s > 0$, and $V_S = \sqrt{\mu/\rho}$ is the transverse wave velocity. $u_p(z)$ is the solution of equation $u_p'' - \eta_p^2 u_p = 0$, where $\eta_p = (\gamma^2 - K_P^2)^{1/2}$, $K_P = (\omega/V_P)$, Re $\eta_p > 0$, and $V_P = ((\lambda + 2\mu)/\rho)^{1/2}$ is the longitudinal wave velocity. Note that, in both cases, $\gamma > K_S > K_P$.

Consider the field in layer $z \in [z_{n-1}, z_n]$ with the constants λ_n, μ_n, and ρ, where $n \in \{1, \ldots, N_z\}$, $z_0 = 0$, $z_{N_z} = -\infty$, and $h_n = z_n - z_{n-1}$ ($n \in \{1, \ldots, N_z - 1\}$) is the layer thickness. According to the general solution of the equation $u'' - \eta^2 u = 0$, in layer n, $u_s(z)$ and $u_p(z)$ can be represented in the form

$$\begin{aligned} u_s(z) &= \overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s(z_n - z_{n-1})} + \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s(z_n - z_{n-1})} \\ u_p(z) &= \overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p(z_n - z_{n-1})} + \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p(z_n - z_{n-1})} \end{aligned} \tag{12}$$

where $\overset{(n)}{A}_s$, $\overset{(n)}{A}_p$, $\overset{(n)}{B}_s$, and $\overset{(n)}{B}_p$ are four constants that depend on the boundary condition and will be determined later.

It follows from (11) and (12) that, in the nth layer, we can obtain the expressions of the displacement and stress tensor

$$\begin{cases} u_x(z) = i\gamma\left(\overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p(z-z_n)} + \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p(z-z_n)}\right) \\ \qquad\quad - \overset{(n)}{\eta}_s\left(\overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s(z-z_n)} - \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s(z-z_n)}\right) \\ u_z(z) = i\gamma\left(\overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s(z-z_n)} + \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s(z-z_n)}\right) \\ \qquad\quad + \overset{(n)}{\eta}_p\left(\overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p(z-z_n)} - \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p(z-z_n)}\right) \\ \sigma_{xz}(z) = \overset{(n)}{Q}_1\left(\overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s(z-z_n)} - \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s(z-z_n)}\right) \\ \qquad\quad + \overset{(n)}{Q}_2\left(\overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p(z-z_n)} - \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p(z-z_n)}\right) \\ \sigma_{zz}(z) = \overset{(n)}{Q}_3\left(\overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s(z-z_n)} - \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s(z-z_n)}\right) \\ \qquad\quad + \overset{(n)}{Q}_4\left(\overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p(z-z_n)} - \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p(z-z_n)}\right) \end{cases} \tag{13}$$

where

$$\begin{cases} \overset{(n)}{Q}_1 = \mu_n\left(-\left(\gamma^2 - \overset{(n)}{\eta}_s{}^2\right)\right) \\ \overset{(n)}{Q}_2 = 2i\gamma\mu_n\overset{(n)}{\eta}_p \\ \overset{(n)}{Q}_3 = 2i\gamma\mu_n\overset{(n)}{\eta}_s \\ \overset{(n)}{Q}_4 = (\lambda_n + 2\mu_n)\overset{(n)}{\eta}_p{}^2 - \lambda_n\gamma^2. \end{cases}$$

According to the upper and lower interfaces of the nth layers in (5), we can derive, together with the impedance tensor [see (7) and (13)], that

$$\begin{cases} u_x(z_{n-1}) - i\gamma\left(\overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p(-h)} + \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p(-h)}\right) \\ \qquad + \overset{(n)}{\eta}_s\left(\overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s(-h)} - \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s(-h)}\right) = 0 \\ u_z(z_{n-1}) - i\gamma\left(\overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s(-h)} + \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s(-h)}\right) \\ \qquad - \overset{(n)}{\eta}_p\left(\overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p(-h)} - \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p(-h)}\right) = 0 \\ \overset{(n)}{Q}_1\left(\overset{(n)}{A}_s - \overset{(n)}{B}_s\right) + \overset{(n)}{Q}_2\left(\overset{(n)}{A}_p - \overset{(n)}{B}_p\right) \\ \qquad - \overset{(n)}{Z}_{xx}(z_n)u_x(z_n) - \overset{(n)}{Z}_{xz}(z_n)u_z(z_n) = 0 \\ \overset{(n)}{Q}_3\left(\overset{(n)}{A}_s - \overset{(n)}{B}_s\right) + \overset{(n)}{Q}_4\left(\overset{(n)}{A}_p - \overset{(n)}{B}_p\right) \\ \qquad - \overset{(n)}{Z}_{zx}(z_n)u_x(z_n) - \overset{(n)}{Z}_{zz}(z_n)u_z(z_n) = 0. \end{cases} \tag{14}$$

For convenience, we rewrite the above system of equations in the form of a matrix equation

$$\begin{aligned} G_n \begin{pmatrix} \overset{(n)}{A}_s \\ \overset{(n)}{B}_s \\ \overset{(n)}{A}_p \\ \overset{(n)}{B}_p \end{pmatrix} &= \begin{pmatrix} \overset{(n)}{G}_{11} & \overset{(n)}{G}_{12} & \overset{(n)}{G}_{13} & \overset{(n)}{G}_{14} \\ \overset{(n)}{G}_{21} & \overset{(n)}{G}_{22} & \overset{(n)}{G}_{23} & \overset{(n)}{G}_{24} \\ \overset{(n)}{G}_{31} & \overset{(n)}{G}_{32} & \overset{(n)}{G}_{33} & \overset{(n)}{G}_{34} \\ \overset{(n)}{G}_{41} & \overset{(n)}{G}_{42} & \overset{(n)}{G}_{43} & \overset{(n)}{G}_{44} \end{pmatrix} \begin{pmatrix} \overset{(n)}{A}_s \\ \overset{(n)}{B}_s \\ \overset{(n)}{A}_p \\ \overset{(n)}{B}_p \end{pmatrix} \\ &= \begin{pmatrix} u_x(z_{n-1}) \\ u_z(z_{n-1}) \\ 0 \\ 0 \end{pmatrix} \end{aligned} \tag{15}$$

where

$$\begin{cases} \overset{(n)}{G}_{11} = \overset{(n)}{\eta}_s e^{-\overset{(n)}{\eta}_s h},\ \overset{(n)}{G}_{12} = -\overset{(n)}{\eta}_s e^{\overset{(n)}{\eta}_s h},\ \overset{(n)}{G}_{13} = -i\gamma e^{-\overset{(n)}{\eta}_p h} \\ \overset{(n)}{G}_{14} = -i\gamma e^{\overset{(n)}{\eta}_p h},\quad \overset{(n)}{G}_{21} = -i\gamma e^{\overset{(n)}{\eta}_s h},\ \overset{(n)}{G}_{22} = -i\gamma e^{\overset{(n)}{\eta}_s h} \\ \overset{(n)}{G}_{23} = -\overset{(n)}{\eta}_p e^{-\overset{(n)}{\eta}_p h},\ \overset{(n)}{G}_{24} = \overset{(n)}{\eta}_p e^{-\overset{(n)}{\eta}_p h} \\ \overset{(n)}{G}_{31} = -\overset{(n)}{Z}_{xx}\overset{(n)}{\eta}_s + i\overset{(n)}{Z}_{xz}\gamma - \overset{(n)}{Q}_1 \\ \overset{(n)}{G}_{32} = \overset{(n)}{Z}_{xx}\overset{(n)}{\eta}_s + i\overset{(n)}{Z}_{xz}\gamma - \overset{(n)}{Q}_1 \\ \overset{(n)}{G}_{33} = -\overset{(n)}{Q}_2 + i\overset{(n)}{Z}_{xx}\gamma + \overset{(n)}{Z}_{xz}\overset{(n)}{\eta}_p \\ \overset{(n)}{G}_{34} = \overset{(n)}{Q}_2 + i\overset{(n)}{Z}_{xx}\gamma - \overset{(n)}{Z}_{xz}\overset{(n)}{\eta}_p \\ \overset{(n)}{G}_{41} = -\overset{(n)}{Z}_{zx}\overset{(n)}{\eta}_s + i\overset{(n)}{Z}_{zz}\gamma - \overset{(n)}{Q}_3 \\ \overset{(n)}{G}_{42} = \overset{(n)}{Z}_{zx}\overset{(n)}{\eta}_s + i\overset{(n)}{Z}_{zz}\gamma + \overset{(n)}{Q}_3 \\ \overset{(n)}{G}_{43} = i\overset{(n)}{Z}_{zx}\gamma + \overset{(n)}{Z}_{zz}\overset{(n)}{\eta}_p - \overset{(n)}{Q}_4 \\ \overset{(n)}{G}_{44} = i\overset{(n)}{Z}_{zx}\gamma - \overset{(n)}{Z}_{zz}\overset{(n)}{\eta}_p - \overset{(n)}{Q}_4. \end{cases} \tag{16}$$

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From the system of linear algebraic equations (15), we can formally obtain

$$\begin{pmatrix} \overset{(n)}{A_s} \\ \overset{(n)}{B_s} \\ \overset{(n)}{A_p} \\ \overset{(n)}{B_p} \end{pmatrix} = G_n^{-1} \begin{pmatrix} u_x(z_{n-1}) \\ u_z(z_{n-1}) \\ 0 \\ 0 \end{pmatrix}. \tag{17}$$

On the other hand, we rewrite the last two equations of (13) in the following vector form:

$$\begin{pmatrix} \sigma_{xz}(z_{n-1}) \\ \sigma_{zz}(z_{n-1}) \end{pmatrix} = \Gamma_n \cdot \Lambda_{\mathrm{n}} \cdot \begin{pmatrix} \overset{(n)}{A_s} \\ \overset{(n)}{B_s} \\ \overset{(n)}{A_p} \\ \overset{(n)}{B_p} \end{pmatrix} \tag{18}$$

where

$$\Lambda_n = \operatorname{diag}\left(e^{\overset{(n)}{\eta_s} h}, \quad e^{-\overset{(n)}{\eta_s} h}, \quad e^{\overset{(n)}{\eta_p} h}, \quad e^{-\overset{(n)}{\eta_p} h}\right)$$

$$\Gamma_n = \begin{pmatrix} \overset{(n)}{Q_1} & -\overset{(n)}{Q_1} & \overset{(n)}{Q_2} & -\overset{(n)}{Q_2} \\ \overset{(n)}{Q_3} & -\overset{(n)}{Q_3} & \overset{(n)}{Q_4} & -\overset{(n)}{Q_4} \end{pmatrix}.$$

Substituting (17) into (18), we obtain

$$\begin{pmatrix} \sigma_{xz}(z_{n-1}) \\ \sigma_{zz}(z_{n-1}) \end{pmatrix} = \Gamma_n \cdot \Lambda_n \cdot G_n^{-1} \begin{pmatrix} u_x(z_{n-1}) \\ u_z(z_{n-1}) \\ 0 \\ 0 \end{pmatrix}. \tag{19}$$

Define

$$(E_n | \hat{E}_n) := \Gamma_n \cdot \Lambda_n \cdot G_n^{-1} \tag{20}$$

where E_n and $\hat{E}_n$ are both matrices of size 2×2. By the definition of E_n, it is a nonlinear function of parameters $\overset{(n)}{Z_{xx}}, \overset{(n)}{Z_{xz}}, \overset{(n)}{Z_{zx}}$, and $\overset{(n)}{Z_{zz}}$ [or equivalently, the matrix $\hat{Z}$ defined in (8)] at the nth layer. That is,

$$E_n = E_n(\hat{Z}_n), \quad \hat{Z}_n = \begin{pmatrix} \overset{(n)}{Z_{xx}} & \overset{(n)}{Z_{xz}} \\ \overset{(n)}{Z_{zx}} & \overset{(n)}{Z_{zz}} \end{pmatrix}. \tag{21}$$

Furthermore, by using (19)–(21), we deduce that

$$\begin{pmatrix} \sigma_{xz}(z_{n-1}) \\ \sigma_{zz}(z_{n-1}) \end{pmatrix} = E_n(\hat{Z}_n) \begin{pmatrix} u_x(z_{n-1}) \\ u_z(z_{n-1}) \end{pmatrix}. \tag{22}$$

From the relation (7), we know that the impedance tensor at the $(n-1)$th layer has the form

$$\begin{pmatrix} \sigma_{xz}(z_{n-1}) \\ \sigma_{zz}(z_{n-1}) \end{pmatrix} = \hat{Z}_{n-1} \begin{pmatrix} u_x(z_{n-1}) \\ u_z(z_{n-1}) \end{pmatrix}. \tag{23}$$

Combine (22) and (23) to obtain $\hat{Z}_{n-1} = E_n(\hat{Z}_n)$, which implies that

$$\hat{Z}_0 = E_1(\hat{Z}_1) = E_1(E_2(\hat{Z}_2)) = \cdots = E_1(E_2(\ldots E_{N_z}(\hat{Z}_{N_z}))). \tag{24}$$

Thus, (10) gives

$$\det E_1(E_2(\ldots E_{N_z}(\hat{Z}_{N_z}))) = 0. \tag{25}$$

TABLE II
EMPIRICAL FORMULAS OF SOME PHYSICAL QUANTITIES

Physical quantity	Calculation formula
μ_n	$\mu_n = 0.466[V_S^{(n)}]^{2.214}$
λ_n	$\lambda_n = 0.466[V_S^{(n)}]^{2.214}$
ρ_n	$\rho_n = 0.466[V_S^{(n)}]^{0.214}$
$V_P^{(n)}$	$V_P^{(n)} = 1.732 V_S^{(n)}$

Note that the quantity $\hat{Z}_{N_z}$ is a matrix-valued function of parameters γ, μ_{N_z}, λ_{N_z}, and ρ_{N_z}. In fact, it follows[1] from [18, Formula (42)] that

$$\hat{Z}_{N_z} = \begin{pmatrix} \overset{(N_z)}{Z_{xx}} & \overset{(N_z)}{Z_{xz}} \\ \overset{(N_z)}{Z_{zx}} & \overset{(N_z)}{Z_{zz}} \end{pmatrix} \tag{26}$$

where

$$\begin{cases} \overset{(N_z)}{Z_{xx}} = -\dfrac{\mu_{N_z} \overset{(N_z)}{\eta_p} \left((\overset{(N_z)}{\eta_s})^2 - \gamma^2 \right)}{\overset{(N_z)(N_z)}{\eta_s \ \eta_p} - \gamma^2} \\ \overset{(N_z)}{Z_{xz}} = \dfrac{i \mu_{N_z} \gamma \left((\overset{(N_z)}{\eta_s})^2 - 2 \overset{(N_z)(N_z)}{\eta_s \ \eta_p} + \gamma^2 \right) \right)}{\overset{(N_z)(N_z)}{\eta_s \ \eta_p} - \gamma^2} \\ \overset{(N_z)}{Z_{zx}} = (\lambda_{N_z} + 2\mu_{N_z}) \dfrac{i\gamma \left(\overset{(N_z)(N_z)}{\eta_s \ \eta_p} - (\overset{(N_z)}{\eta_p})^2 \right)}{\overset{(N_z)(N_z)}{\eta_s \ \eta_p} - \gamma^2} + i\gamma \lambda_{N_z} \\ \overset{(N_z)}{Z_{zz}} = (\lambda_{N_z} + 2\mu_{N_z}) \dfrac{\overset{(N_z)}{\eta_s} \left((\overset{(N_z)}{\eta_p})^2 - \gamma^2 \right)}{\overset{(N_z)(N_z)}{\eta_s \ \eta_p} - \gamma^2}. \end{cases} \tag{27}$$

For simplicity, we assume that $h_n \equiv h$ for all $n = 1, \ldots, N_z - 1$. Now, let us discuss the values of parameters μ_n, λ_n, and ρ_n in the nth layer. To that end, denote by $V_S^{(n)}$ and $V_P^{(n)}$ the S-wave velocity and the P-wave velocity in the nth layer, respectively. By the empirical formula (cf. [43]), we have $\rho_n = 0.466[V_S^{(n)}]^{0.214}$. Moreover, it is well known that (see [7]) $V_S^{(n)} = (\mu_n/\rho_n)^{1/2}$ and $V_P^{(n)} = ((\lambda_n + 2\mu_n)/\rho_n)^{1/2}$, which implies, together with the empirical formula $V_P^{(n)} = 1.732 V_S^{(n)}$ (see [44]) and the mentioned relation $\rho_n = 0.466[V_S^{(n)}]^{0.214}$, that $\lambda_n = \mu_n = 0.466[V_S^{(n)}]^{2.214}$. To summarize, in this work, μ_n, λ_n, and ρ_n are purely dependent on the value of the S-wave velocity $V_S^{(n)}$ (see Table II).

In practice, the phase velocity C of the Rayleigh wave at a given frequency ω is usually measured. It is well known that $C = \omega/\gamma$ ([7]). To that end, denote by $\bar{V}_S = (V_S^{(1)}, \ldots, V_S^{(N_z)})$ the vector of the S-wave velocity in our layered medium model of the Earth. By the construction of matrix $\{E_n\}_{n=1}^{N_z}$, we conclude that the left-hand side in (25) is a nonlinear continuous function of the parameters C and $\bar{V}_S$. According to the implicit function theorem, we deduce from (25) that, in some neighborhood of the point $\bar{V}_S$, there exists a differentiable function f of $\bar{V}_S$ such that

$$f(\bar{V}_S) = C. \tag{28}$$

[1]This can be obtained by using (3), (7), and the vanishing boundary conditions (6).

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TABLE III
THREE-LAYER MEDIUM MODEL OF THE EARTH

Layer	h	ρ	V_P	V_S
1	10	2.4	3.5	2.1
2	20	2.5	4	2.7
3	∞	3	6	4

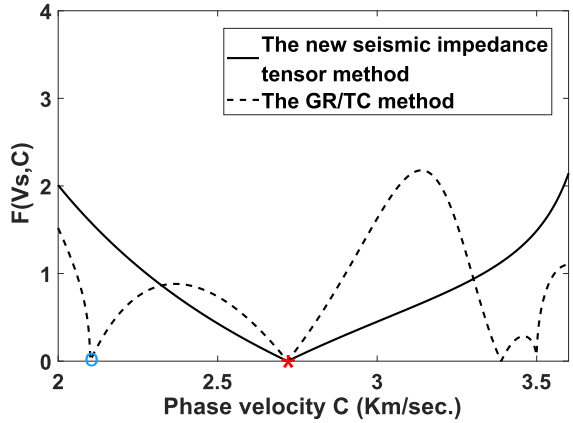


Fig. 2. Solutions for C provided by an existing method (GR/TC method) and our new method. The function $F(V_S, C)$ denotes the left-hand side of (25) at a given $T = 20$ s.

Hence, for a given S-wave velocity $\bar{V}_S$, one can efficiently compute the phase velocity C through the procedure (28). It should be noted that, although we do not obtain the closed form of function f, the value of C can be numerically determined by solving a nonlinear equation (25). Since both the physical quantity C and the function f depend on the frequency, given a frequency $\{\omega_m\}_{m=1}^{N_\omega}$, one can obtain a sequence of equations

$$f(\bar{V}_S; \omega_m) = C(\omega_m), \quad m = 1, \ldots, N_\omega.$$

Define $\mathbf{f} = (f(\bar{V}_S; \omega_1), \ldots, f(\bar{V}_S; \omega_{N_\omega}))$ and $\mathbf{C} = (C(\omega_1), \ldots, C(\omega_{N_\omega}))$. Without ambiguity, we can call $\mathbf{C}$ the dispersion curve (in fact, $\mathbf{C}$ represents the discretized version of the dispersion curve). Now, we obtain the vector form of the equation

$$\mathbf{f}(\bar{V}_S) = \mathbf{C}. \tag{29}$$

At the end of this section, we provide a numerical example to demonstrate that our new model (29) can avoid the pitfall of "mistake solutions" that arise in classical methods, e.g., the generalized reflection–transmission coefficient (GR/TC) method. The simulation is based on the three-layer medium model specified in Table III. The results are displayed in Fig. 2, from which we can conclude that the newly introduced Rayleigh wave dispersion function $\mathbf{f}$ has only a single root, thereby avoiding the nonuniqueness issue encountered in the original inverse problem.

III. MACHINE LEARNING-BASED INVERSION SOLVER

This section is devoted to the development of an efficient machine learning method to estimate the S-wave velocity $\bar{V}_S$ at a given frequency from the dispersion curve and our new mathematical model (29). In this inversion solver, we assume that both $\bar{V}_S \in \mathbb{R}^{N_z}$ and $\mathbf{C} \in \mathbb{R}^{N_\omega}$ are random vectors, where N_z and N_ω are the total numbers of crustal layers and frequencies, respectively. The problem of estimating S-wave velocity $\bar{V}_S$ from the measured dispersion curve $\mathbf{C}$ is transformed into estimating the function from $\mathbf{C}$ to $\bar{V}_S$, which is solved by using the models based on neural networks in this work.

TABLE IV
RANGES FOR RANDOMLY GENERATING ENTRIES IN $\mathbf{x}$

	3-layer	5-layer	9-layer
$[a_0, b_0]$	[3.00, 4.00]	[3.00, 3.80]	[3.00, 3.80]
$[a_1, b_1]$	[3.80, 4.80]	[3.20, 4.00]	[3.10, 3.90]
$[a_2, b_2]$	[4.60, 5.60]	[3.80, 4.60]	[3.20, 3.95]
$[a_3, b_3]$	-	[3.80, 4.60]	[3.30, 4.00]
$[a_4, b_4]$	-	[4.00, 4.80]	[3.80, 4.60]
$[a_5, b_5]$	-	-	[3.90, 4.70]
$[a_6, b_6]$	-	-	[4.00, 4.75]
$[a_7, b_7]$	-	-	[4.20, 4.80]
$[a_8, b_8]$	-	-	[4.60, 5.60]

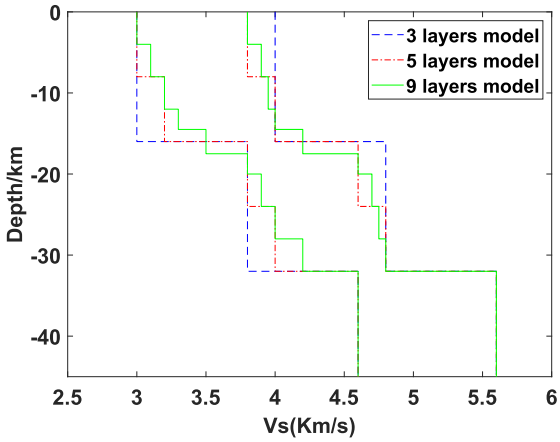


Fig. 3. Control range of V_S velocity structure in three media models (three, five, and nine layers).

A. Data Structure

We use neural networks to address the three-, five-, and nine-layer cases (i.e., $N_z = 3, 5, 9$) separately. Let $(\mathbf{x}_j, \mathbf{y}_j) := (\{x_{lj}\}_{l=0}^{N_z-1}, \{y_{mj}\}_{m=0}^{N_\omega-1})$ represent the jth realization of $(\bar{V}_S, \mathbf{C})$ in the N_z-layer case, where $N_\omega = 50$ in all of our simulations. The N samples[2] are generated in the following way.

1) For each of $l \in \{0, 1, \ldots, N_z - 1\}$ and $j \in \{0, 1, \ldots, N - 1\}$, x_{lj} is sampled independently of Uniform (a_l, b_l), where a_l and b_l are taken from the global crustal average model ([45]), the values of which are listed in Table IV. This choice of x_{lj} reflects the control range of velocities V_S structure in the three media models (three, five, and nine layers) (see Fig. 3).
2) The 50 entries of $\mathbf{y}_j$ are computed by using the forward model (29) with the generated $\mathbf{x}_j$. Note that $\{h, \boldsymbol{\omega}_m\}$ are also involved in the computation process,

[2]We generate 48 000 models for the three-layer case and the five-layer case, and generate 120 000 models for the nine-layer case, as it is more complicated.

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whose values are preselected in the following way. The thickness of each layer is assumed to be equal, and h is taken as 16 km for the three-layer case, 8 km for the five-layer case, and 4 km for the nine-layer case. To obtain the dense samples for the dispersion curve $\mathbf{y}$ (or equivalently, $\mathbf{C}$), according to experience, in this work, the frequencies $\boldsymbol{\omega}_m$ for the data collection are designed quasi-uniformly according to the formula $\boldsymbol{\omega}_m = 2\pi\{0.5^{1/399} + (80^{1/399} - 0.5^{1/399})(m-1)/49\}^{-399}$, $m = 1, \ldots, N_\omega = 50$. Other parameters, such as V_P and ρ, are defined in Table II.

Moreover, in the simulation, we then randomly assign 80% of the N samples for model training, 10% for validation, and 10% for testing.

B. Addressing the Inversion Problem With a Machine Learning Method Based on Mixture Density Networks

As demonstrated in Appendix A, due to the nonuniqueness of the inversion mapping of (29), the ordinary FNNs are not suitable for the considered inverse problem. In this section, we address this problem by using the MDN.

1) Brief Introduction to the MDN: Suppose that the dataset $\{(\mathbf{x}_j, \mathbf{y}_j)\}_{j=0}^{N-1}$ is generated by an unknown multivalued function $f^{-1}(\mathbf{y})$ with K branches $\{f_k^{-1}(\mathbf{y})\}_{k=0}^{K-1}$. Then, it is intuitive to characterize the data with a probability model. For example, for each sample $(\mathbf{x}, \mathbf{y})$, it can be assumed that

$$\mathbf{x} = \sum_{k=0}^{K-1} \pi_k f_k^{-1}(\mathbf{y})$$

where $\pi_k \in [0, 1]$ is the probability that $(\mathbf{x}, \mathbf{y})$ lies on the kth branch.

The mixture model is based on a similar idea, but, rather than deterministic values, it allows $\mathbf{x}$ on the kth branch (for $k \in \{0, 1, 2, \ldots, K-1\}$) to be the realization of a certain distribution, e.g., a Gaussian distribution with mean $f_k^{-1}(\mathbf{y})$. Specifically, for any sample $(\mathbf{x}, \mathbf{y})$, the Gaussian mixture (GM) model assumes that $\mathbf{x}$ is drawn from a conditional distribution $\boldsymbol{X}|\boldsymbol{Y} = \mathbf{y}$ with the following probability density function (pdf):

$$p_{\boldsymbol{X}|\boldsymbol{Y}=\mathbf{y}}(\mathbf{x}) = \sum_{k=0}^{K-1} \pi_k p_k(\mathbf{x}) \tag{30}$$

where p_k denotes the pdf of a Gaussian distribution $\mathcal{N}(\mu_k, \sigma_k^2)$ and the weight $\pi_k \in [0, 1]$ satisfies $\sum_{k=0}^{K-1} \pi_k = 1$. Since all the parameters' values depend on $\mathbf{y}$, we can also write them as $\pi_k(\mathbf{y}), \mu_k(\mathbf{y}), \sigma_k^2(\mathbf{y})$. Note that the kth branch $f_k^{-1}(\mathbf{y})$ in the above deterministic case corresponds to $\mathcal{N}(\mu_k(\mathbf{y}), \sigma_k^2(\mathbf{y}))$.

The MDN, when the prior is chosen to be Gaussian, is a GM with one more assumption: $\pi_k(\mathbf{y})$, $\mu_k(\mathbf{y})$, and $\sigma_k^2(\mathbf{y})$ are continuous functions and, hence, can be approximated with an FNN with input $\mathbf{y}$.

2) MDN Model Structure: The structure of an MDN is the same as that of an FNN whose input is $\mathbf{y}$ and whose target output is a GM distribution, which can be determined by the parameters $\{\pi_k(\mathbf{y}), \mu_k(\mathbf{y}), \sigma_k^2(\mathbf{y})\}_{k=0}^{K-1}$. To be more precise, suppose that the dimensionality of $\mathbf{x}$ is n; then, μ_k and σ_k also share the same dimensionality. Thus, the dimensionality of (μ_k, σ_k, π_k) is $2n+1$. Hence, the number of nodes in the output layer of the FNN is $(2n+1)K$. We index nodes in the output layer with $\{0, 1, \ldots, (2n+1)K-1\}$ and divide them into K sets. Then, we set the target of the kth set as $(\pi_k(\mathbf{y}), \mu_k(\mathbf{y}), \sigma_k^2(\mathbf{y}))$. More specifically, let $\{o_j\}_{j=0}^{(2n+1)K-1}$ denote the FNN output for an input $\mathbf{y}$. Then, the following hold.

1) The target value for $\hat{\mu}_k(\mathbf{y}) := [o_{(2n+1)k}, o_{(2n+1)k+1}, \ldots, o_{(2n+1)k+(n-1)}]$ is $\mu_k(\mathbf{y})$, where a ReLU activation is used for these nodes to make sure the outputs are nonnegative.
2) The target value for $\hat{\sigma}_k(\mathbf{y}) := [o_{(2n+1)k+n}, o_{(2n+1)k+n+1}, \ldots, o_{(2n+1)k+(2n-1)}]$ is $\sigma_k(\mathbf{y})$, where a sigmoid activation multiplied by a positive hyperparameter[3] is used, and
3) $o_{\pi,k} := o_{(2n+1)k+2n}$ is put through a softmax function to approximate $\pi_k(\mathbf{y})$. Specifically, we construct

$$\hat{\pi}_k := \frac{e^{o_{\pi,k}}}{\sum_{l=0}^{K-1} e^{o_{\pi,l}}}, \quad k = 0, 1, \ldots, K-1$$

the target values of which are $\pi_k(\mathbf{x})$. The point of using a softmax function is to ensure that $\hat{\pi}_k \in [0, 1]$ and $\sum_{k=0}^{K-1} \hat{\pi}_k = 1$.

Note that, in practice, the target values $\{\mu_k(\mathbf{y}), \sigma_k(\mathbf{y}), \pi_k(\mathbf{y})\}_{k=0}^{K-1}$ are not known. Thus, when constructing the loss function, the MDN uses the maximum likelihood method. Maximum likelihood estimation is used to set the loss for training the MDN. It is set as the negative of the estimated log likelihood of the data. Specifically,

$$\begin{aligned} \text{MDN Loss} &= -\sum_{j=0}^{N-1} \log(\hat{p}_{\boldsymbol{X}|\boldsymbol{Y}=\mathbf{y}_j}(\mathbf{x}_j)) \\ &= -\sum_{j=0}^{N-1} \log\left(\sum_{k=1}^{K} \hat{\pi}_k(\mathbf{y}_j)\hat{p}_k(\mathbf{x}_j)\right) \end{aligned} \tag{31}$$

where $\hat{p}_k$ denotes the pdf of $\mathcal{N}(\hat{\mu}_k(\mathbf{y}_j), \hat{\sigma}_k^2(\mathbf{y}_j))$. Therefore, minimizing the loss is equivalent to looking for weights in the FNN structure of the MDN that maximizes the log likelihood of the dataset.

Based on the above setting, at the end of this section, we construct a new model based on the MDN. Our new model, named FW-MDN, consists of an MDN with a modified loss function that consists of two parts. One is the ordinary MDN loss, and the other is the distance between $\hat{\mathbf{y}}$ and $\mathbf{y}$, where $\hat{\mathbf{y}}$ is the prediction made by the forward model using the MDN outputs. Specifically, our loss function is

$$\begin{aligned} \text{Loss} := -\sum_{j=0}^{N-1} \log(\hat{p}_{\boldsymbol{X}|\boldsymbol{Y}=\mathbf{y}_j}(\mathbf{x}_j)) + \sum_{j=0}^{N-1} |\hat{\mathbf{y}}_j - \mathbf{y}_j|^2 \\ + \alpha_b \sum_{l=0}^{B-1} b_l^2 + \alpha_w \sum_{l=0}^{W-1} w_l^2 \end{aligned} \tag{32}$$

[3]This hyperparameter is set to adapt to variances in the data samples, and its value is determined by the validation-set approach. We find, in our later experiments in Table V, that its optimal value is 0.001 among the candidate set of $\{1, 0.1, 0.01, 0.001, 0.0001\}$.

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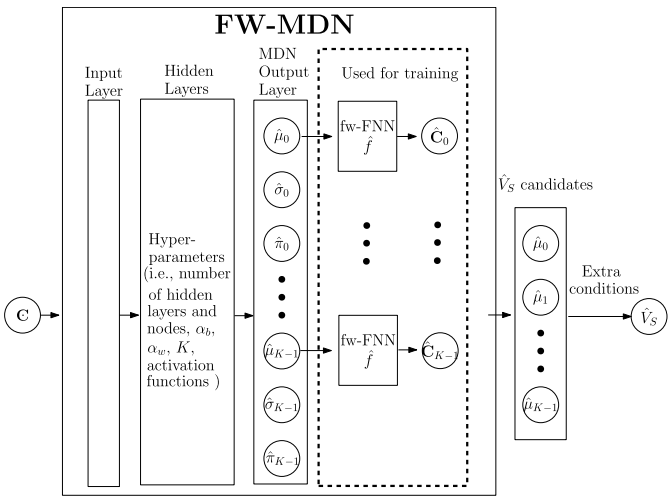


Fig. 4. FW-MDN structure and the overall inversion procedure.

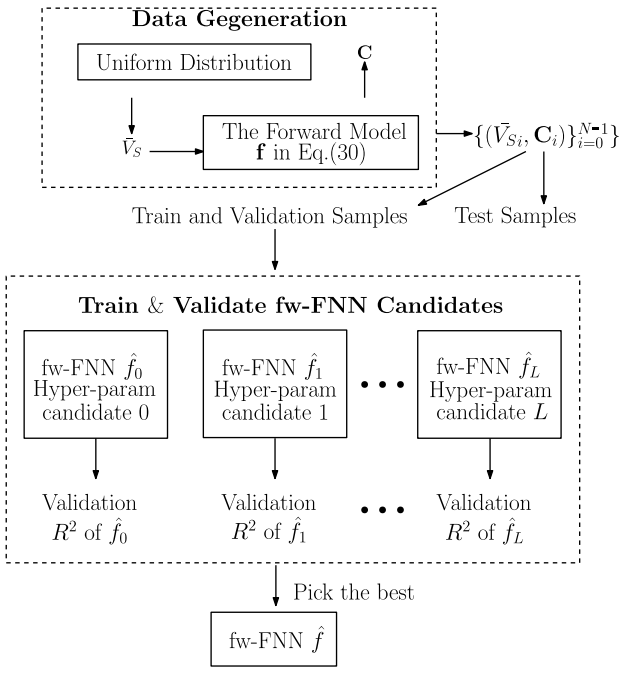


Fig. 5. Data generation and the fw-FNN included in an MDN.

where the first negative sum is the MDN loss defined in (31). In the second sum, $\hat{\mathbf{y}}_j = \sum_{k=0}^{K-1} \pi_k \hat{f}(\hat{\mu}_k(\mathbf{y}_j))$, where $\{\hat{\mu}_k\}$ are part of the MDN outputs (see Fig. 4) and $\hat{f}$ is an approximation[4] of the forward model $\mathbf{f}$ in (29). The third and fourth sums are regularizations for the bias terms $\{b_l\}_{l=0}^{B-1}$ and weights $\{w_l\}_{l=0}^{W-1}$ associated with the input, hidden, and MDN output layers. Moreover, α_b and α_w are the regularization coefficients, which are treated as hyperparameters.

For each sample input $\mathbf{y}$ (a realized value of $\mathbf{C}$), the trained FW-MDN returns part of the MDN output, $\{\hat{\mu}_k\}_{k=1}^{K}$, as a set of K possible values for $\bar{V}_S$. Although no extra conditions are imposed by the FW-MDN to identify the "correct" one among these values, at least one of them should be close[5] to the true $\bar{V}_S$ value with a high probability, under the assumption that $\bar{V}_S$ follows the GM distribution in (30). This point is verified by our later results in Tables VI and VII. A visualization of the FW-MDN can be found in Fig. 4, where fw-FNN (see Fig. 5) denotes the pretrained FNN, which is also the factor $\hat{f}$ used in the loss, which approximates the forward model $\mathbf{f}$.

The trained FW-MDN is "physics-informed" because the physics information (i.e., the mapping rule from S-wave velocities $\bar{V}_S$ to Rayleigh wave phase velocities $\mathbf{C}$) encoded in the forward physical model $\mathbf{f}$ in (29) is implicitly used when training the FW-MDN.

The fw-FNN $\hat{f}$ is trained on the artificial datum $\mathscr{D}$ generated by the forward physical model $\mathbf{f}$ in (29), and it approximates $\mathbf{f}$ well, with a validation R^2 above 99%. Therefore, the mapping rule (physical law) from $\bar{V}_S$ to $\mathbf{C}$ is also encoded in $\hat{f}$. This rule is implicitly used when training the FW-MDN because both $\mathscr{D}$ and $\hat{f}$ are used in the training process. Specifically, $\hat{f}$ is used to compute a term in the FW-MDN loss [see (32)], which is $\sum_{j=0}^{N-1} |\hat{\mathbf{y}}_j - \mathbf{y}_j|^2$, where j represents the jth sample and $\hat{\mathbf{y}}_j := \sum_{k=0}^{K-1} \hat{\pi}_k(\mathbf{y}_j)\hat{f}(\hat{\mu}_k(\mathbf{y}_j))$. Intuitively, including this term in the loss causes the FW-MDN to produce $\bar{V}_S$ estimates that match (according to the mapping rule) the input $\mathbf{C}$ values.

[4]The forward model $\mathbf{f}$ is approximated with an FNN using the training and validation sample sets. Its validation R^2 is above 99%. Recall that $\mathbf{f}$ is used for sample generation.

[5]This is because μ_k is the mode of the kth component of the GM distribution for $k \in \{0, 1, 2, \ldots, K-1\}$.

TABLE V

SELECTED VALUES FOR THE HYPERPARAMETERS AND ACTIVATIONS. α_b IS THE REGULARIZATION COEFFICIENT FOR BIAS TERMS, AND α_w IS THE REGULARIZATION COEFFICIENT FOR WEIGHTS

Model	Hidden Layers and Nodes	Activation
fw-FNN	(50)	ReLU
FW-MDN	(100, 50)	tanh

Model	α_b	α_w	K
fw-FNN	0.001	0.001	NA
FW-MDN	0.001	0.00001	4

C. Training and Testing of the FW-MDN

Training the FW-MDN is, in fact, training the involved FNN.[6] Note that fw-FNN is pretrained, and its weights do not change when training the FW-MDN. The model hyperparameters are selected using the well-known validation-set approach[7] and are shown in Table V. Appendix B discusses the effect of a key hyperparameter, K, on FW-MDN's performance.

Since each sample $(\mathbf{x}_j, \mathbf{y}_j)$ at most lies on one branch of the multivalued function, it can match only one element in the FW-MDN output set $\{\hat{\mu}_k\}_{k=0}^{K-1}$. Hence, we construct the measure for model performances as the distance between $\mathbf{x}_i$

[6]Only the weights associated with the "Input Layer" and "Hidden Layers" in Fig. 4 are trained when training the FW-MDN.

[7]We preselect a set of candidates, where each candidate consists of values for all hyperparameters (including K) in the FW-MDN. Then, we train an FW-MDN for each candidate and pick the one whose corresponding model's validation performance is the best. Please refer to [46, Sec. 5.1.1] for more details on this method.

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TABLE VI
FW-MDN TEST PERFORMANCES

Entry	Test $\mathscr{M}$	Entry	Test $\mathscr{M}$
x_0	99.9%	x_5	88.0%
x_1	99.8%	x_6	85.9%
x_2	99.1%	x_7	90.4%
x_3	94.9%	x_8	99.9%
x_4	86.2%	Overall	93.8%

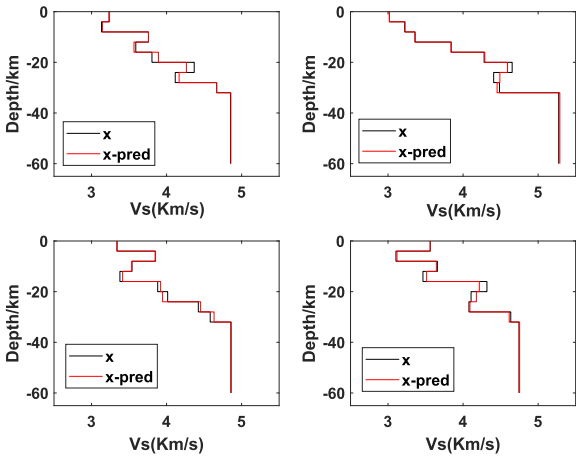


Fig. 6. Predicted $\hat{\mu}^*$ and the true $\bar{V}_s$ (i.e., $\mathbf{x}$) for four randomly chosen samples.

and its closest neighbor in $\{\hat{\mu}_k(\mathbf{y}_j)\}_{k=0}^{K-1}$. That is, we define the performance measure statistic as

$$\mathscr{M} = 1 - \frac{\sum_{i=0}^{N-1} \left(\hat{\mu}_j^* - \mathbf{x}_j\right)^2}{\sum_{j=0}^{N-1} (\mathbf{x}_j - \bar{\mathbf{x}})^2}$$

where $\bar{\mathbf{x}} := (1/N)\sum_{j=0}^{N-1} \mathbf{x}_j$ and $\hat{u}_j^* := \arg\min_{\hat{\mu}_k(\mathbf{y}_j), k\in\{0,\ldots,K-1\}} |\hat{\mu}_k(\mathbf{y}_j) - \mathbf{x}_j|^2$. The construction of $\mathscr{M}$ is similar to that of the commonly used measure for model performance R^2, and the closer $\mathscr{M}$ is to 1, the better performance of the FW-MDN. This measure reflects the capacity of the FW-MDN to include the true $\mathbf{x}_j$ in its outputs, which is appropriate for any model that approximates a multivalued function. We call $\hat{\mu}^*$ the FW-MDN prediction for convenience although the FW-MDN, in fact, predicts a set $\{\hat{\mu}_k\}_{k=0}^{K-1}$.

The testing performances of the trained FW-MDN are reported in Table VI. We use the TensorFlow library[8] [47] for model training and testing. Compared to the ordinary FNN, whose testing R^2 values are reported in Table IX, there is a 25.8% increase in the FW-MDN's overall performance, and its scores in the middle layers, e.g., $\{x_5, x_6, x_7\}$, are improved significantly, from below 40% to above 80%.[9] The predictions $\hat{\mu}^*$ for four randomly chosen samples are plotted[10] in Fig. 6.

[8]https://tensorflow.google.cn/?hl=en

[9]Although the two measures are different ($\mathscr{M}$ is for a multivalued function, and R^2 is for a single-valued function), they are comparable since their calculating formulas are similar.

[10]Figs. 6 and 8 are plotted with MATLAB, while Figs. 7 and 9 are plotted with Python.

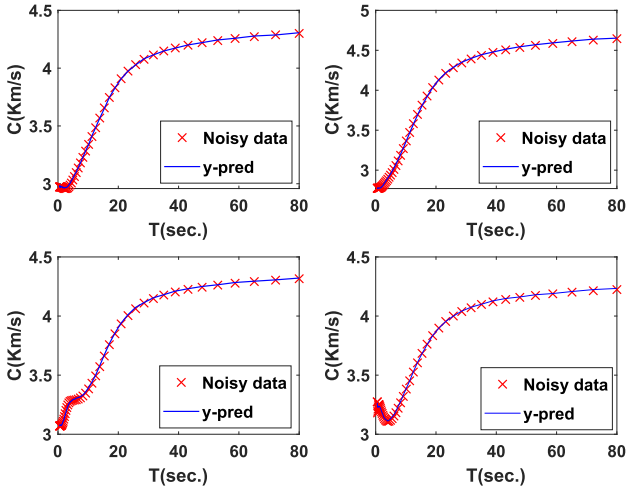


Fig. 7. Predicted dispersion curves ($\hat{\mathbf{y}}$) and the true values ($\mathbf{y}$) for four randomly chosen samples. The horizontal axis represents T.

To see whether the FW-MDN prediction $\hat{\mu}^*$ produces dispersion curves close to the true values (i.e., the true $\hat{\mathbf{y}}$), on the test sample set, we compute the R^2 between $\{\hat{f}(\hat{\mu}_j^*)\}_{j=0}^{N-1}$ and $\{\mathbf{y}_j\}_{j=0}^{N-1}$, and obtain a satisfying result of 94.8%, indicating that our FW-MDN prediction and the true value of $\bar{V}_s$ produce dispersion curves that are close to each other. Fig. 7 plots $\hat{f}(\hat{\mu}^*)$ and $\mathbf{y}$ for the four samples in Fig. 6.

D. Robustness of the FW-MDN Under Artificial Noise

Because measurement errors in the dispersion curves are common, the model's robustness against noise needs to be investigated. Hence, we test the previously trained FW-MDN's performance on artificially noised testing samples.

Specifically, for each $(\mathbf{x}_j, \mathbf{y}_j)$ in the test set, define

$$\mathbf{y}_j^{\text{noised}} := \mathbf{y}_j(1 + \varepsilon_j) \tag{33}$$

where $\{\varepsilon_i\}$ are uniformly (and independently) drawn from $(-0.5\%, 0.5\%)$. Then, we input $\{\mathbf{y}_j^{\text{noised}}\}$ into the FW-MDN and compute $\mathscr{M}$ between $\{\mathbf{x}_j\}$ and their nearest neighbors $\{\hat{\mu}_j^*\}$ in the FW-MDN outputs. The result is -2.95, which indicates that the previously trained FW-MDN is not robust against measurement errors.[11]

To create models that are robust against noise, we build new FW-MDNs,[12] with the training and validation samples artificially noised with uniformly distributed errors (the same as in formula (33)), which enables the model to learn to discern noises. Specifically, when training the model, we input the noised data into the model, but the losses (32) are computed with the model outputs $\{\hat{\mathbf{y}}_j\}$ and unnoised data $\{(\mathbf{x}_j, \mathbf{y}_j)\}$.

To determine how much noise should be added to the training dataset, for each of the noise types $\{\text{Unif}(-0.2\%, 0.2\%)$, $\text{Unif}(-0.5\%, 0.5\%)$, $\text{Unif}(-0.8\%, 0.8\%)$, $\text{Unif}(-1\%, 1\%)\}$, we train an FW-MDN (using the hyperparameters in Table V),

[11]If the noise drawn from $N(0, 0.2\%^2)$ is added to the test data, the test performance of the FW-MDN becomes -1.12, which also implies nonrobustness against noise.

[12]We keep the model structures and hyperparameters the same as those in Table V.

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TABLE VII
FW-MDN PERFORMANCE $\mathscr{M}$ ON NOISED TEST SAMPLES

Noise added for training and validation: U(−0.8%,0.8%)					
Target	Noise-free	$N(0, 0.25\%^2)$	$N(0, 0.50\%^2)$	U(−0.50%, 0.50%)	U(−1%, 1%)
x	80.5%	79.3%	75.8%	78.8%	74.1%
y	99.4%	99.3%	99.1%	99.3%	99.0%
$\mathbf{y}^{noised}$	NA	99.0%	98.0%	98.9%	97.5%
x_0	99.9%	99.8%	99.7%	99.8%	99.7%
x_1	99.3%	99.0%	97.8%	98.8%	97.4%
x_2	93.2%	90.9%	84.1%	90.0%	81.6%
x_3	78.0%	75.5%	68.8%	74.7%	66.4%
x_4	74.6%	73.6%	70.0%	73.2%	68.6%
x_5	74.9%	74.5%	73.3%	74.3%	72.2%
x_6	62.2%	60.6%	54.7%	59.5%	53.2%
x_7	42.2%	39.5%	33.8%	38.9%	28.1%
x_8	99.9%	99.9%	99.7%	99.8%	99.7%

Using the validation-set approach, we select hyperparameter values and train an FW-MDN. Its hyperparameter values are the same as those in Table V, except for the bias regularization coefficient, which is 0.00001. Each column records the performance of the FW-MDN on test data noised with a certain error type. For example, in the column $N(0\%, 0.5\%^2)$, we independently draw $\varepsilon_i \sim N(0\%, 0.5\%^2)$ and noise the test data following (33). Then, we compute $\mathscr{M}$ between the test data $\{\mathbf{x}_i\}$ and the FW-MDN outputs. The row **x** reports the overall $\mathscr{M}$. The rows x_l, $l = 0, 1, \cdots, 8$, report $\mathscr{M}$ for the lth entry of **x**. The row **y** reports the R^2 between **y** and $\hat{f}(\hat{\mu}^*)$. The row $\mathbf{y}^{noised}$ reports the R^2 between $\mathbf{y}^{noised}$ and $\hat{f}(\hat{\mu}^*)$. U denotes a uniform distribution.

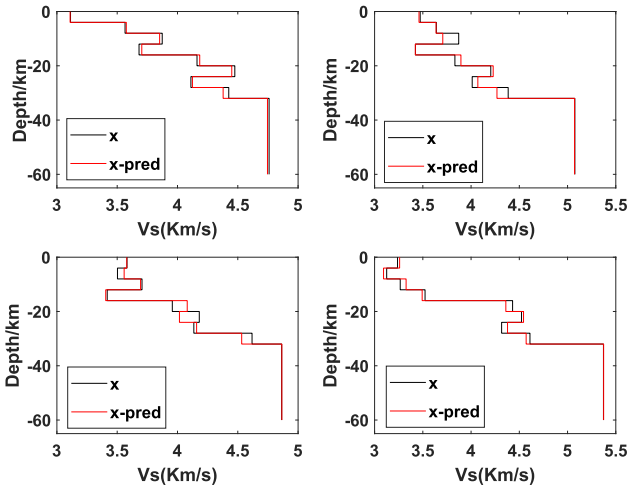


Fig. 8. Prediction $\hat{\mu}^*$ (i.e., $\mathbf{x} - \mathbf{pred}$) and the true $\bar{V}_s$ (i.e., **x**) for four randomly chosen noised test samples.

and we select the one with the best performance on the validation data noised with errors uniformly distributed in $(-1\%, 1\%)$. The selected noise is Unif$(-0.8\%, 0.8\%)$, with which we noise the training and validation data, and build a new FW-MDN. In Table VII, we report its performance in testing samples noised with various types of artificial errors. The results show that, compared with the FW-MDN trained with noise-free data, the newly trained FW-MDN is much more robust against noise. For example, the overall $\mathscr{M}$ values obtained on test samples with various types of noise are all above 70%, which are satisfactory in most real-world cases. Fig. 8 plots $\hat{\mu}^*$ obtained from the model and **x** for four randomly drawn testing samples that are noised with Unif$(-1\%, 1\%)$ errors, from which we can see that the predictions $\hat{\mu}^*$ are close to the true values of **x**. Fig. 9 plots $\mathbf{y}^{noised}$ and $\hat{f}(\hat{\mu}^*)$ for the four samples in Fig. 8, in which the noised

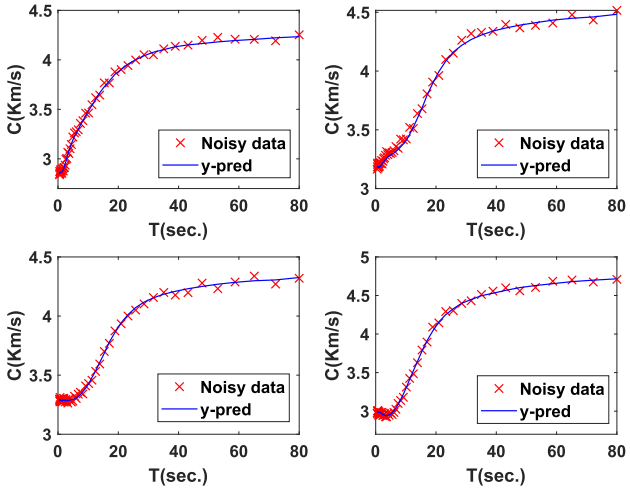


Fig. 9. Plots of $\mathbf{y}^{noised}$ (noisy data) and $\hat{f}(\hat{\mu}^*)$ ($\mathbf{y} - \mathbf{pred}$) for the four samples in Fig. 8.

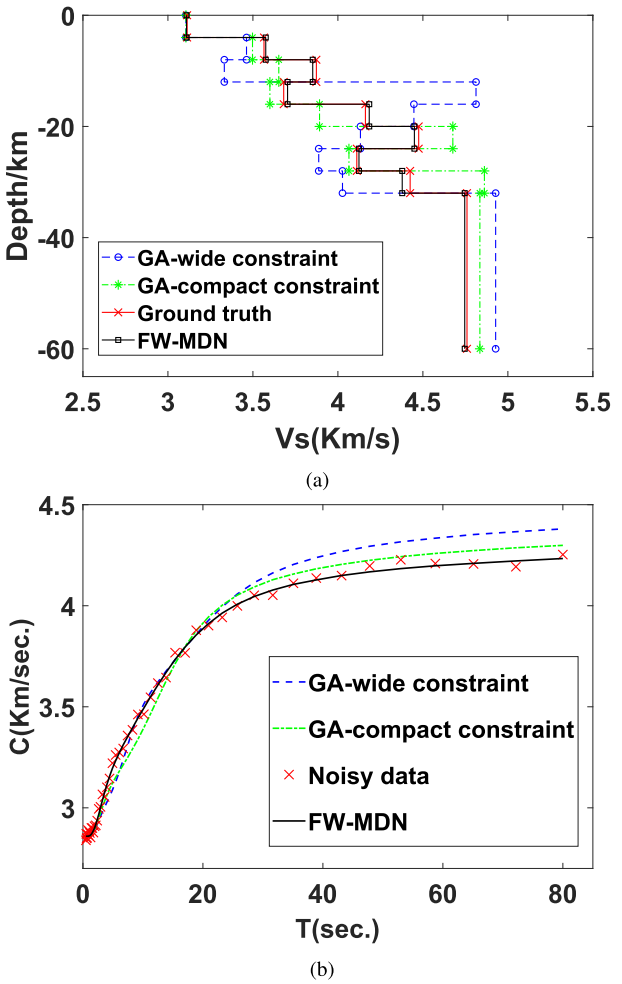


Fig. 10. Comparisons with a GA (population size: 100) with a wide or compact constraint. (a) Estimated $\bar{V}_s$. (b) Predicted C.

dispersion curves **y** are also close to the predictions $\hat{f}(\hat{\mu}^*)$ obtained by forwardly propagating the FW-MDN outputs. All these results show that the newly trained FW-MDN is robust against noise.

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To conclude this section, the FW-MDN is compared with a GA. The results are displayed in Fig. 10, from which one can see that the FW-MDN's predictions are clearly closer to the true values. It should be noted that, once the FW-MDN has been trained, the computational complexity is low for each prediction, whereas, for the GA, the computation required for each prediction is much heavier and takes a long time. In our example, the GA takes more than 5 h, while the FW-MDN needs less than a second for each prediction.

IV. Conclusion

In this work, we develop a new Rayleigh wave dispersion equation (28) [or (29)], from which the dispersion curve can be easily computed. Based on this new forward model, we use the structure of an MDN to develop a new inversion solver, named FW-MDN, whose training loss function involves the forward model, to reconstruct the S-wave velocity from the data of dispersion curves. The FW-MDN is capable of dealing with the nonuniqueness phenomenon that appeared in our inverse geophysics problem, which other existing studies that apply machine learning fail to accommodate.

Although more experience with real-world data is needed to fully understand the potential and the limitations of the FW-MDN, the initial applications to the model problem and the artificial data system are promising, yielding excellent fits of the data and a well-defined S-wave velocity. Moreover, the proposed novel forward model and the machine learning-based inversion solver can easily be extended to the case with a Rayleigh wave multiple-mode dispersion curve. We, therefore, believe that the FW-MDN will be useful in the study of many inverse problems in geophysics.

Appendix A
Performances of Ordinary Feedforward Neural Networks and Their Failure in a Multilayer Medium Model

In this appendix, we address the inversion problems (29) with ordinary FNNs. For each case, we build an ordinary FNN to predict the S-wave velocity $\bar{V}_S$ from the dispersion curve $\mathbf{C}$. The hyperparameters, such as the numbers of hidden layers and nodes, regularization coefficients, and activation functions, are selected using the validation-set approach. The performance of the optimal FNN in each case is listed in Table VIII, which shows that the FNN can solve the inversion problem perfectly for the three- and five-layer cases. However, its performance decreases sharply for the nine-layer case. Table IX reports the performance of the FNN on each entry of the input vector, from which we can see that the predictions for the middle layer V_S are off. For example, R^2 is merely 30.7% for V_S of the eighth layer (x_7).

Now, let us discuss the reasons for the FNN underperformance for the case of the nine-layer medium model of the Earth. In fact, among our simulated nine-layer data, we find that certain data samples have very close $\mathbf{y}$ values, yet their $\mathbf{x}$ values differ significantly. Fig. 11 illustrates this phenomenon with two samples from our dataset. This is the well-known nonuniqueness problem. It implies that the target function

TABLE VIII

Model Performances With Different Combinations of Hyperparameters and Activations. $R^2 = 1 - ((\sum_{j=0}^{N-1}(\hat{\mathbf{x}}_j - \mathbf{x}_j)^2)/(\sum_{i=0}^{N-1}(\mathbf{x}_j - \bar{\mathbf{x}})^2))$, Where $\bar{\mathbf{x}} := (1/N)\sum_{j=0}^{N-1}\mathbf{x}_j$. α_b Is the Regularization Coefficient for Bias Terms, and α_w Is the Regularization Coefficient for Weights

Case	Hidden Layers and Nodes		Activation
3-layer	(150, 150, 150, 100, 100)		tanh
5-layer	(150, 150, 150, 100, 100)		tanh
9-layer	(400, 400, 300, 300, 300)		tanh
α_b	α_w	Train R^2	Test R^2
0.001	0.001	99.7%	99.7%
0.001	0.001	98.7%	98.7%
0.001	0.001	66.5%	66.4%

TABLE IX

FNN Test Performances

Entry	Test R^2	Entry	Test R^2
x_0	98.3%	x_5	29.5%
x_1	96.7%	x_6	37.2%
x_2	81.2%	x_7	30.7%
x_3	48.2%	x_8	99.3%
x_4	37.4%	Overall	66.5%

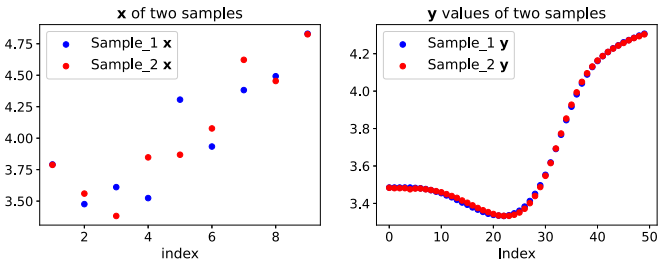


Fig. 11. Plots of two samples in the artificial dataset of our problem.

for our learning model is a multivalued function,[13] which an ordinary FNN is not able to approximate. This is because an FNN is a single-valued function. A recognized learning model that can handle this approximation problem is the MDN, which is introduced in Section III-B1.

Appendix B
Effect of K's Value on FW-MDN's Performance

The value of K significantly affects the performance of the FW-MDN. In theory, as the value of K increases (while other hyperparameters are held fixed at their optimal values), the performance of the trained FW-MDN (newly built and trained for each value of K) on the validation samples should

[13] Suppose that f is a "noninjective" mapping from a set $\mathscr{X}$ to another set $\mathscr{Y}$, i.e., for a certain $\mathbf{y} \in \mathscr{Y}$; there are $\{\mathbf{x}_k\}_{k=0}^{K-1} \subset \mathscr{Y}$ such that $f(\mathbf{x}_k) = \mathbf{y}$. Then, the inverse mapping, denoted by f^{-1}, is called a *multivalued* function. If there also exist K single-valued functions $\{f_k^{-1}(\mathbf{y})\}_{k=0}^{K-1}$ such that $f_k^{-1}(\mathbf{y}) = \mathbf{x}_k$, then each g_k is called a *branch* of f^{-1}. A multivalued function in fact refers to a relation $F(\mathbf{x}, \mathbf{y}) = 0$ such that, for a certain value of $\mathbf{y}$, e.g., $\mathbf{y}_0$, there is more than one value of $\mathbf{x}$, e.g., $\{\mathbf{x}_0, \mathbf{x}_1\}$, such that $F(\mathbf{x}_k, \mathbf{y}_0) = 0$ for $k = 0, 1$.

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TABLE X

VALIDATION SCORES OF FOUR FW-MDNS WITH DIFFERENT K VALUES. THE OTHER HYPERPARAMETERS ARE THE SAME AS IN TABLE V. THE MEASURE $\mathscr{M}$ IS DEFINED IN SECTION III-C. THE FW-MDNS ARE TRAINED ON NOISE-FREE SAMPLES

K	Validation $\mathscr{M}$
1	83.7%
2	91.7%
3	93.0%
4	93.8%

improve significantly until K reaches the number of branches of the multivalued function $\mathbf{f}^{-1}$ that is being approximated. This pattern is observed when we apply the validation-set approach to select the hyperparameter values. For example, as K ranges from $\{1, 2, 3, 4\}$, while other hyperparameters are held fixed at the values reported in Table V, the performance of the FW-MDN on the validation samples increases from 83.7% to 91.7% and then remains at the 90% level, as shown in Table X.

These results indicate that the number of branches of the target multivalued function is at least 2.

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